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Project Title:	The Contribution of Delta Rays to the Chromosome Aberration Dose Response in Human Cells Irradiated with HZE Particles of Different Energy but the Same LET		
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Zip Code:	77555-5302	**Congressional District:**	14
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	Highly energetic heavy ions (HZE), of the type found in galactic cosmic rays, deposit energy in the form of ionizations along the tracks they take as they travel through a cell. These ionizations are tightly spaced along these tracks and this spacing makes them highly damaging to cellular structures. Some of the particles' energy is also imparted to electrons (delta rays) which can travel great distances and produce damage, albeit less severe, in cells far removed from the original particle track. The ranges of these electrons are proportional to the energy of the original HZE particle. While these delta rays by themselves are not likely to produce much in the way of measurable damage, we have evidence that these electrons can add damage to that forming along independent HZE particle tracks that will eventually be expressed in the form of chromosome exchanges of greater complexity. Chromosome exchanges result when radiation breaks a chromosome by severing the DNA of which it is composed. The cell has means to repair the chromosome and can splice		

Task Description:	the DNA together. On some occasions, if one chromosome break is close to another break, the cell might make a mistake and join the broken ends to inappropriate partners. Two or more breaks can be involved in the process; the greater the number of breaks, the more complicated the resulting chromosome exchange will be. We plan to test the hypothesis that these exchange complexity increases will be related to the primary particle energy and resultant maximum delta ray range. Higher particle energies will result in longer delta ray maximum ranges allowing more primary particle tracks to be close enough to irradiate a cell with delta electrons for any specific fluence (particle tracks per unit area; equivalent to dose). We plan to irradiate human skin cells with low and high energy beams matching LET (linear energy transfer; energy deposited per unit track length) and fluences so that the only variable will be particle energy. A dose response curve for chromosome exchanges irradiated with low particle fluences will be constructed. At the same time we will also look for breaks that fail to rejoin. One experiment will be conducted at low dose rate which will prevent damage from delta rays from being added to chromosome exchanges forming along primary particle tracks by spreading damage formation out in time and allowing some breaks to be repaired or exchanges to be completely formed before other breaks occurring later can be added to them. We also plan to grow populations from single surviving cells allowing us to measure the transmissibility of chromosome exchanges a number of cell generations removed from the irradiation. Chromosome aberrations are tightly linked to cancer development. With this study we hope to gain a better understanding of the processes underlying chromosome exchange formation following irradiation with HZE particles similar to those astronauts might encounter in space. We plan to determine the shape of the exchange dose response curve for these particles at low doses and at low dose rate, processes involving the misrepair of radiation induced damage that could lead to carcinogenesis. At the same time we will look at terminal deletions and incomplete exchanges resulting from the cell's failure to repair damage. These lesions are likely to kill the cell and eliminate it from a population that might develop into cancer. Finally, we will look for exchanges found only in a fraction of the cells making up a clonal population indicating genomic instability, the process which drives carcinogenesis. This project will endeavor to gain understanding on how delta ray interactions with primary particle tracks might affect these endpoints that are associated with processes that lead to cancer development and enable NASA to produce more accurate models for cancer risk estimates from exposure to galactic cosmic rays.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overarching goal of this proposal is to determine the effect of a single charged particle track passing through a cell nucleus without any influence from d-rays contributed by independent tracks passing through the medium at the same time. This is the condition likely to occur in space and understanding the effects produced under these conditions is important in order to assess the dangers associated with galactic cosmic rays (GCR). Regrettably, simulating these conditions in the Earth environment is virtually impossible as even at the smallest doses, effects of a single track will be impacted by d-rays originating from independent tracks. It is the goal of this study to measure these track interaction effects on the chromosome exchange breakpoint dose response in order to separate them from those produced by single primary tracks. A thorough understanding of the mechanisms that underlie track interaction should allow the construction of more precise biophysical models of the effects of HZE particles on living organisms thereby reducing the uncertainty in cancer risk estimates NASA seeks in order to assess the hazards of GCR. In addition, track interaction is likely to have effects on particle beam therapy giving this study further societal and economic impact.
Task Progress:	Last year we began our study using low energy (1.8 MeV/n) 7Li ions from the Brookhaven National Laboratory (BNL) Tandem Van de Graaff accelerator to irradiate lymphocytes and fibroblasts and measure the resulting chromosome aberrations produced in these cells in their first post-irradiation mitosis. While we had intended to start with high energy ions at the NASA Space Radiation Laboratory (NSRL) we were informed that the Tandem was likely to close soon and it was suggested that we complete our experiments there as soon as possible. This was a rather unfortunate circumstance as the facility, at that time, was not equipped to irradiate a large number of live cells under the sterile conditions necessary for our study. We were aware that studies of the type we wanted to conduct were performed by investigators from Columbia University's Radiological Accelerator Facility (RARAF) several years ago. We discussed with the Columbia group how they performed their experiments and were able to borrow some of the apparatus they had used. The key problem with low energy particle irradiations is that their range after leaving the vacuum of the accelerator is very short. This feature of the beam requires a thin window (2.5 μm thick Havar) at the end of the beam line located as close as possible to the dish containing the cells and the use of a thin Mylar film (6 μm thick) as the surface of the dish on which the cells are attached. Several preliminary experiments were conducted in the Principal Investgator's laboratory at the University of Texas Medical Branch (UTMB) to determine the best procedures for handling the cells and isolating the necessary mitotic cells needed for chromosomal analysis. In spite of all our preparation, the first experiment did not go well and we were unable to produce any usable data. We were able to ship some irradiated cells back to UTMB where we have generated clonal populations deriving from single cells that survived the treatment. This was done to determine the transmissibility of chromosomal aberrations to future cell generations; such aberrations being potential initiating events in carcinogenesis. We were able to isolate 19 clones each from the 0.4 and 1.2 Gy irradiated populations. Chromosomal analysis of these clones is ongoing. Since our last report, we made modifications to our procedures and repeated the 7-Li experiment in November of last year. While the experiment did not go as well as we would have liked, we were able to obtain some usable data. The resulting data set is rather limited but is not inconsistent with the linear dose response expected for these low-energy high linear energy transfer (LET) particles, although more data are needed to say this conclusively. Chromosome aberrations are formed when radiation or some other treatment breaks chromosomes. Normally, the cell has the means to repair these breaks and joins the chromosomes back together. On occasion, if a number of breaks are close to one another, the cell can make a mistake and join the end one side of a break (break end) on one chromosome to one on another chromosome while the other break end from the first chromosome often is joined to the remaining break end on the second chromosome. This leads to an exchange of chromosomal segments that can be detected with chromosome analysis procedures. While many times, as in this example, exchanges involve only two breaks (termed simple exchanges), other times three or more breaks may be involved in an exchange. For example, a break end from chromosome "A" can join to one from chromosome "B" while the other side of the break from "B" can join to one of the ends of a break on chromosome "C". The remaining break end from chromosome "C" can join to the open break end from chromosome "A" or it may join to a break end from on a fourth chromosome and so on. When three or more breaks are involved, the exchange is described as a "complex exchange." When we examined our chromosome aberration data, we discovered that the lymphocytes had a higher frequency of

	chromosome breaks as indicated by the presence chromosomal exchanges than did the fibroblasts and this increased frequency was associated with breaks forming complex exchanges. When a high LET particle travels through a cell it places the ionizations that ultimately cause chromosome breaks quite close to one another favoring the production of chromosome exchanges, particularly complex ones. The differences we saw in the frequencies of chromosome breakpoints are likely due to differences in the geometry of cell nuclei between the two cell types. The lymphocyte has a spherical nucleus with a diameter of 6 μm (mean chord length of 4 μm) while the nucleus of a fibroblast can best be described as a flat plate with a thickness of about 1 μm. Therefore, a particle traveling through a lymphocyte nucleus will travel a greater distance and have more opportunities to produce chromosome breaks than one traveling through the thinner fibroblast nucleus. This may, in part, explain the differences observed in the breakpoint frequencies of the two cell types. In the spring, we conducted our first experiment at NSRL using 1 GeV/n 56-Fe ions. To improve our yields of mitotic cells we decided to irradiate whole blood cultures. As a result, the lymphocyte arm of the experiment was quite successful. The fibroblast arm once again proved problematic with low yields of mitotic cells, although significantly improved over the yields from the 7Li experiments. Chromosomal analysis is ongoing for this experiment and we should have reportable results soon. We were also able to generate 24 clonal populations for transmissibility analysis. Last week, we conducted another experiment at NSRL using 91.7 MeV/n 28-Si ions. We modified our procedures once again and it would appear this might be our best experiment yet. We made the slides while at BNL, which improved the quality of our preparations. Yields seem quite good with the exception of the higher doses with the fibroblasts. We should be able to begin chromosomal analysis in the next few weeks. We have also taken steps with live cells to isolate clonal populations for transmissibility analysis.
Bibliography Type:	Description: (Last Updated: 10/29/2023)
Abstracts for Journals and Proceedings	Loucas BD, Cornforth MN. "Damage interaction produced by independent HZE particles at low fluence. I. Low energy 7-Li ions." Presented at the 61st Annual Meeting of the Radiation Research Society, Weston, FL, September 19-22, 2015. 61st Annual Meeting of the Radiation Research Society, Weston, FL, September 19-22, 2015. Conference proceedings distributed at the meeting. , Sep-2015
Abstracts for Journals and Proceedings	Loucas BD, Cornforth MN. "The production of chromosome exchanges by low energy 7Li ions: A base line for track interaction of HZE particles in the chromsome exchange breakpoint dose response." 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. Conference proceedings distributed at the meeting. , Jan-2015